

Beyond the Grade Book

Spotting students at risk of failing Portuguese — before it is too late

[Team name / Group number] · [Member names]

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One in seven students fails Portuguese

15%

of 649 students failed the course (final grade below 10 out of 20)

2

public secondary schools in Portugal (GP and MS), school year 2005–06

Too late

schools usually learn who is struggling only when final grades are released

Why it matters: Portuguese is a core subject. Failing it blocks progression and is a strong early sign of leaving school. Support staff and tutoring time are limited, so schools need to know where to focus them — early.

Three questions for school leaders

1

What lowers final grades?

Which things a school knows at enrolment go with a higher or lower final grade?

2

Who is at risk of failing?

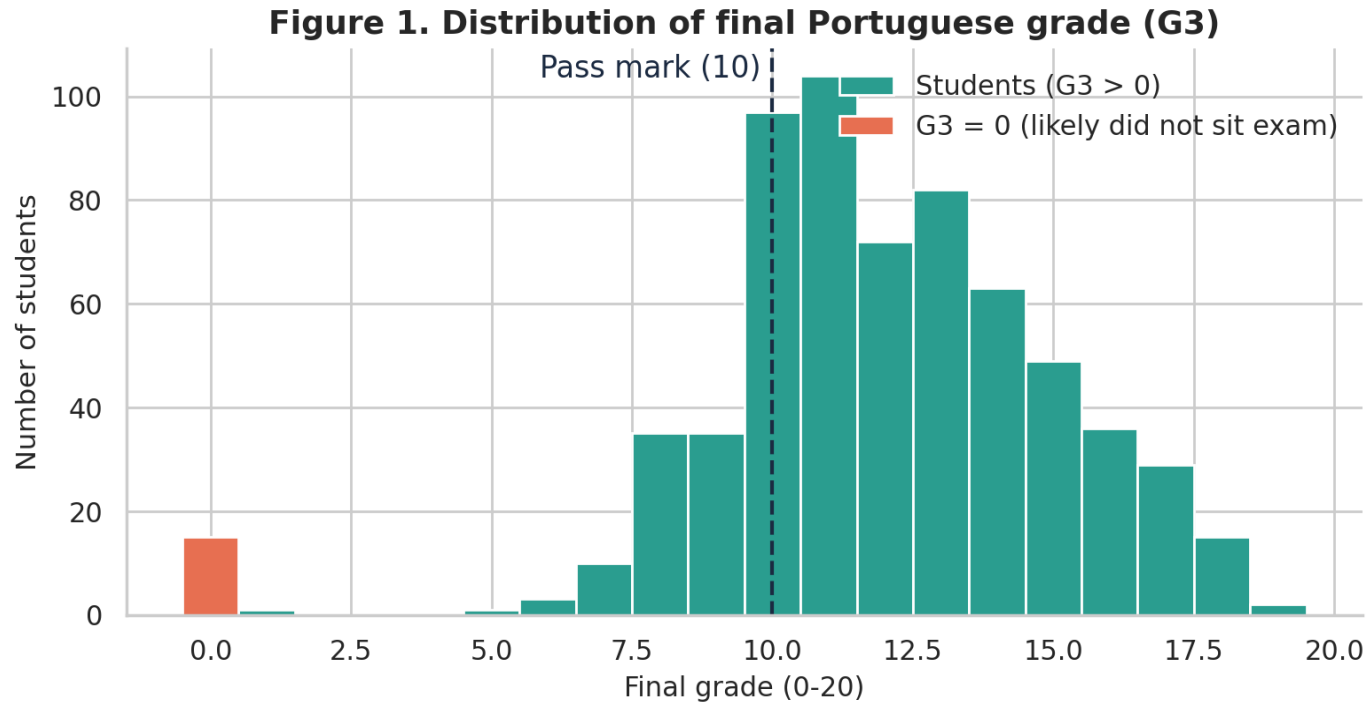
Can the same information pick out the students most likely to fail?

3

When is the warning reliable?

How much better is the warning after the first-term grade is known?

The data: 649 students, 33 pieces of information each



Most students score 10–14 out of 20. The pass mark is 10.

Source

Public school records and student questionnaires
(Cortez & Silva, 2008; Kaggle, CC0)

Quality

No missing values, no duplicates, all values in valid ranges

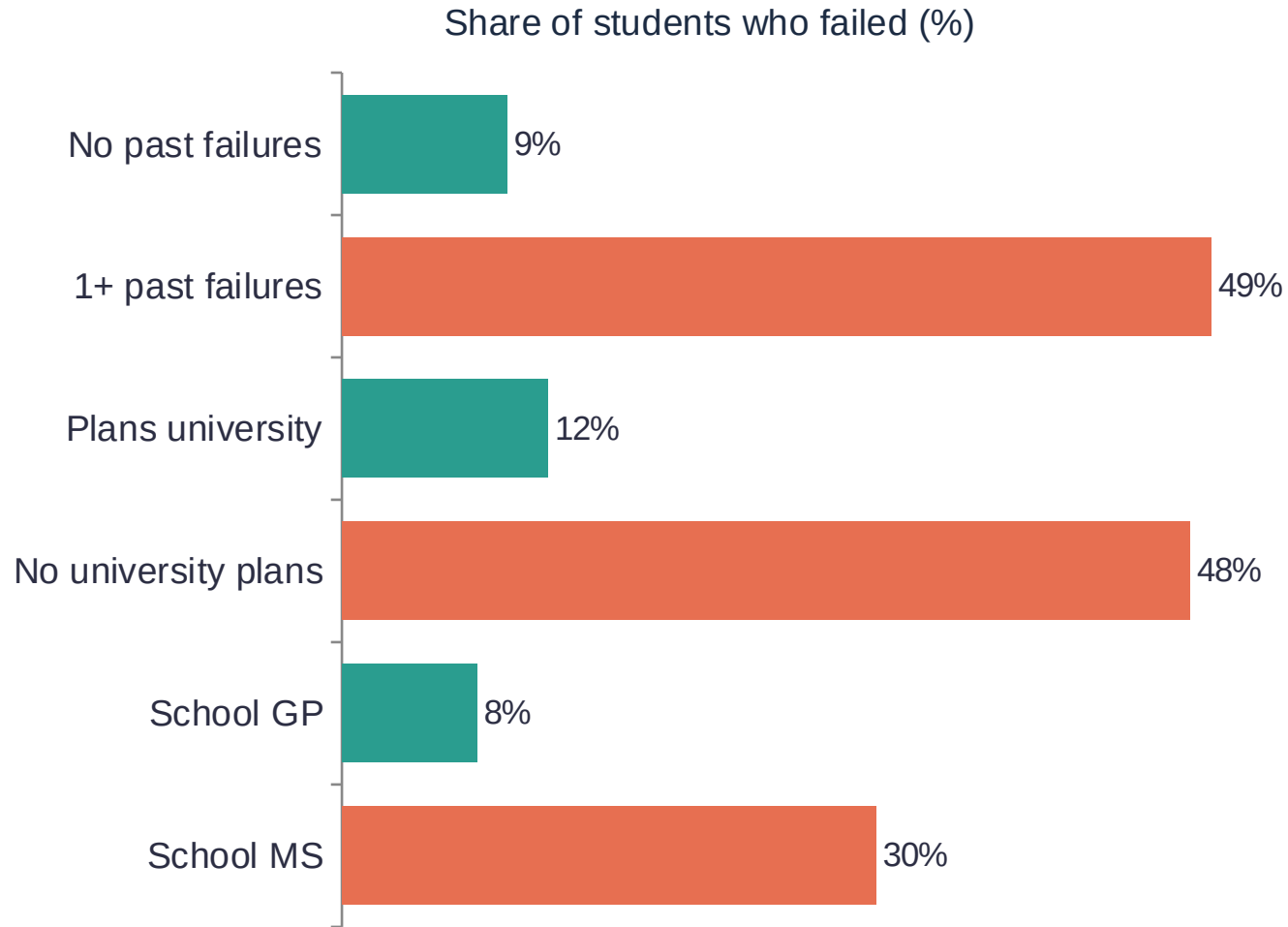
One issue

15 students scored 0 but had no absences — they most likely did not sit the final exam

Our decision

Left out of grade models, but counted as 'fail' in the risk model

Three warning signs stand out



Past failures

Students who failed a class before are 5× more likely to fail again (49% vs 9%).

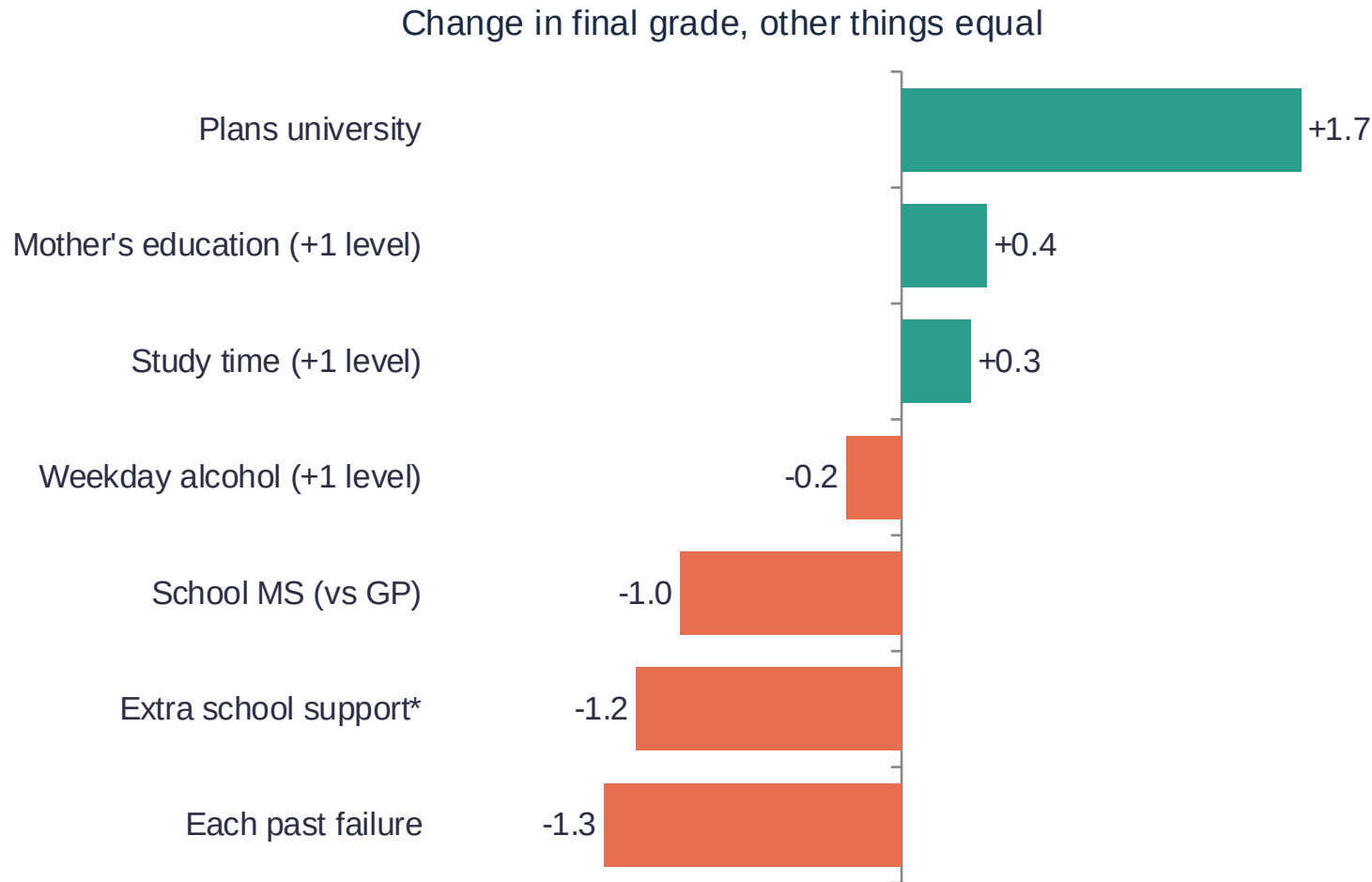
University plans

Students without plans for higher education fail 4× as often (48% vs 12%).

School

MS students fail 4× as often as GP students (30% vs 8%).

What moves the final grade? (out of 20 points)



About one third

of the differences in grades are explained by enrolment information alone.

* **Support does not cause lower grades** — weaker students are the ones who receive it. This is a link, not a cause.

Model: multiple linear regression, 634 students, 12 factors; all shown effects are statistically significant.

Can we spot who will fail? Yes — fairly well

84%

chance the model ranks a student who fails above one who passes (AUC 0.84)

3 in 4

students who fail would be flagged at enrolment (74%)

1 in 3

students would be flagged in total — a manageable list for support staff

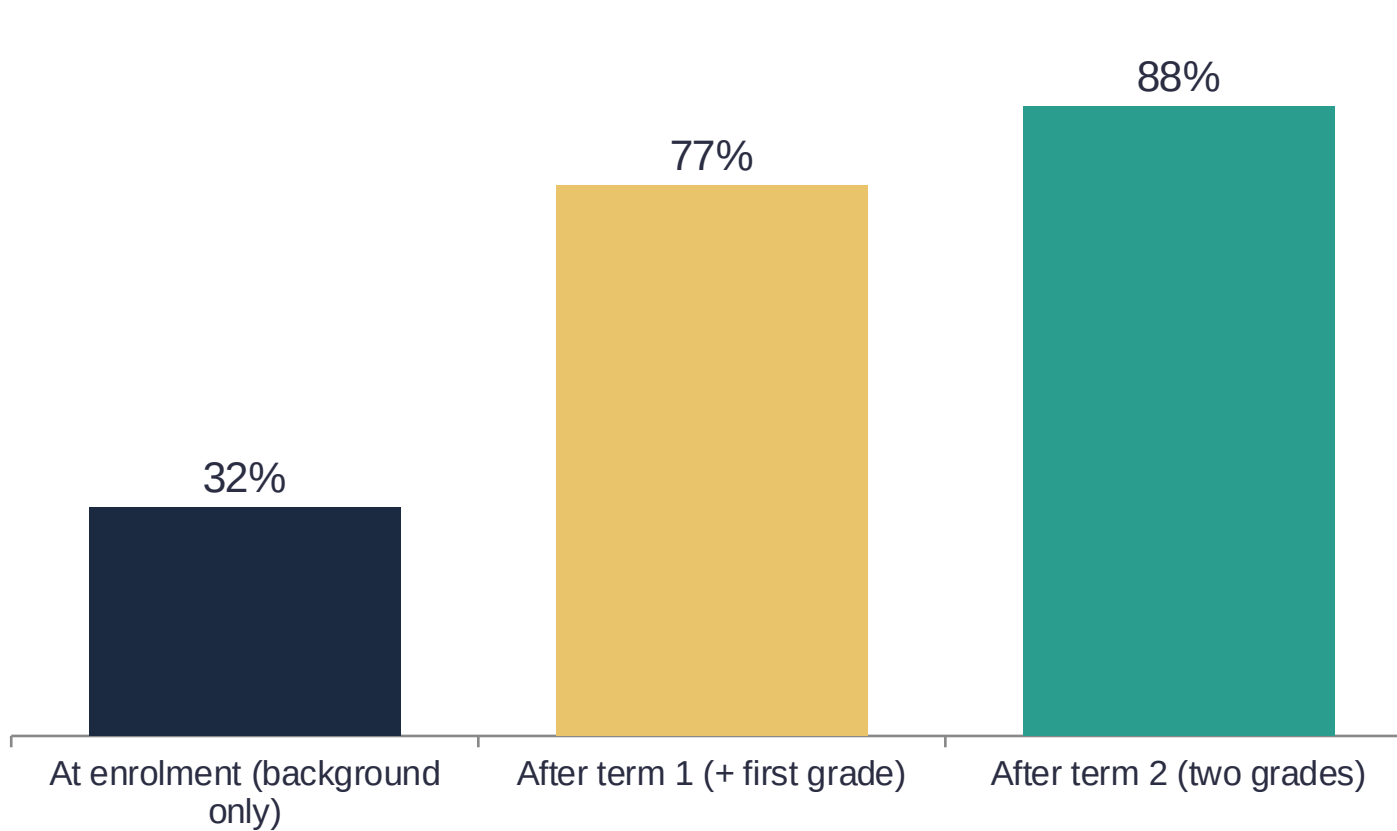
Biggest risk factors (logistic regression, 649 students)

School MS: odds of failing about 7× higher · Each past failure: odds 3× higher · Each extra absence: +7%

Planning to go to university: odds almost 4× lower (protective)

The first-term grade makes the warning much sharper

Share of grade differences predicted (tested on unseen students)



± 1.7

points typical error at enrolment

± 1.0

points after term 1

± 0.7

points after term 2

Message: start with a background check at enrolment, then re-check after the first term.

Can we trust these results?



Tested on unseen students

10-fold cross-validation: results hold on data the model did not see



Model checks passed

Straight-line patterns, no overlapping factors ($VIF \leq 1.2$), no single student drives results



Protected against uneven spread

Used robust standard errors (HC3), so conclusions stay valid



Some caution

Grade is a whole number 0–20, so residuals are slightly non-normal; large sample limits the impact

What our study cannot tell you

- 1 Link $\neq$ cause**

We observed patterns. We cannot say that changing one factor will change grades.
- 2 Old, local data**

Two schools in one region of Portugal, 2005–06. Other schools or today's students may differ.
- 3 Self-reported answers**

Study time and alcohol use come from questionnaires, so students may under- or over-report.
- 4 Small fail group**

Only 100 students failed, so the risk model should be re-checked on new data before use.

Our recommendations

1

Flag at enrolment

Use past failures, university plans and school to build a first support list.

2

Re-check after term 1

Update the list with first-term grades — accuracy more than doubles.

3

Support, don't label

Use the list to start conversations and offer tutoring, never to limit a student's options.

Ethics: any future data about Aboriginal and Torres Strait Islander students should follow Indigenous data sovereignty (CARE, Maiam nayri Wingara): co-design with community, collective consent, community control and strengths-based reporting.

Thank you

Questions?

Data: Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. FUBUTEC 2008. Kaggle mirror: Student Performance Data Set (CC0).